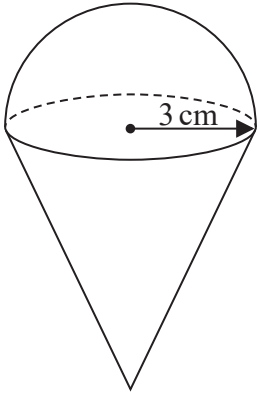
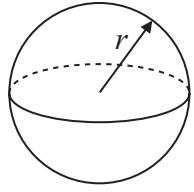
19 The diagram shows a solid made by joining a solid hemisphere to a solid circular cone.

The centre of the plane face of the cone coincides with the centre of the plane face of the hemisphere.



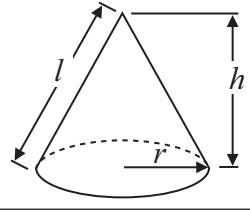
Surface area of sphere = $4\pi r^2$

Volume of sphere = $\frac{4}{3}\pi r^3$



Curved surface area of cone = πrl

Volume of cone = $\frac{1}{3}\pi r^2 h$



The radius of the hemisphere is 3 cm.
The radius of the base of the cone is 3 cm.

The volume of the solid is 30π cm^3

Work out the total surface area of the solid.
Give your answer as a multiple of π.

..... cm^2

(Total for Question 19 is 5 marks)